

Advanced Java – Introduction

Assignment

Objective

The purpose of this assignment is to introduce the concepts, architecture, and commonly used technologies in Advanced Java. Complete the questions in your own words and include suitable examples where required.

Part A – Short Answer Questions

1. What is Advanced Java? How is it different from Core Java?
2. Explain the role of JDBC in Java application development.
3. What is a Servlet? Mention two advantages of using Servlets.
4. What is JSP and how does it differ from a Servlet?
5. Explain the Model–View–Controller (MVC) architecture in Java web applications.
6. What is a web container? Give one example.
7. What is a JavaBean? List its important characteristics.
8. Explain the purpose of the HTTP GET and POST methods.
9. What is session management? Name two techniques used for maintaining sessions.
10. What is the purpose of the JDBC DriverManager class?

Part B – Descriptive Questions

11. Explain the steps involved in connecting a Java application to a database using JDBC.
12. Describe the Servlet life cycle with the functions of `init()`, `service()`, and `destroy()`.
13. Explain the JSP life cycle and the relationship between JSP and Servlets.
14. Describe cookies, URL rewriting, hidden form fields, and `HttpSession` as session-tracking techniques.
15. Explain the advantages of using MVC architecture in Java web development.

Part C – Practical Tasks

16. Create a simple Java program that establishes a JDBC connection to a database and displays a successful connection message.
17. Create a basic Servlet that accepts a user's name from an HTML form and displays a welcome message.
18. Create a simple JSP page that displays student details using JSP expressions or script-free Expression Language.
19. Design a small MVC-based Java web application concept for managing student records. Identify the Model, View, and Controller components.

Submission Requirements

- Write clear and concise answers.
- For practical tasks, submit source code and screenshots of the output.
- Use meaningful class, variable, and method names.
- Mention the database and server/container used for JDBC or web tasks.
- Test all programs before submission.

Topics Covered

Topic	Key Concepts
Advanced Java	Java web and enterprise application basics
JDBC	Database connectivity, drivers, statements, ResultSet
Servlets	Life cycle, request/response handling
JSP	Dynamic web pages, JSP elements, EL
MVC	Separation of Model, View, and Controller
Session Management	Cookies, HttpSession, URL rewriting